MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-403( )/TSC-76

1. SCOPE

1.1 Scope.- This specification covers a transportable shelter facility which is part of and incorporates all facilities for the installation and operation of components of Communicating Patch Panel AN/TSC-76( ) for use within the Army Area Communications System.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-6858	Welding Resistance
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-55505	Shelter, Electrical Equipment (containing equipment), Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-22	Welded - Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Mechanical and Visual Defects

DRAWINGS

DL-SC-A-595476	Supplemental Data List, Shelter Electrical Equipment S-403( )/TSC-76
SC-E-595477	Signal Schematic S-403( )/TSC-76

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(Copies of Specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description.- Shelter, Electrical Equipment S-403( )/TSC-76 (hereafter referred to as shelter facility) is a modified S-250( )/G Shelter, which, upon the installation of certain operating equipment becomes as AN/TSC-76( ). The AN/TSC-76( ) is a versatile mobile switching unit capable of terminating and interconnecting 1144 local circuits. Test facilities are provided for testing all local circuits. One (1) TT-98( )/FG is provided for monitoring teletypewriter circuits. Cable connectors and binding posts are provided for making connections to other components in the area communications center. The shelter facility requires 115 VAC 60 cycles single phase primary power.

3.2 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the S-403( )/TSC-76 for approval. (See 4.3)

3.3 Operational Equipment.- Where specified herein, the following government loaned equipment shall be installed in the shelter facility to be tested, unless otherwise specified in the bid request or contract.

1 each Switchboard, Telephone Manual SB-22( )/PT
1 each Intercommunication Station LS-147C/FI
1 each Telephone Set TA-312( )/PT
1 each Terminal Telegraph TH-22( )/TG
1 each Teletypewriter TT-98/FG
1 each Headset, Microphone, H-144( )/U

3.4 Construction.- The shelter facility shall be constructed in accordance with the model supplied, the list of exceptions to the model, schematic SC-E-595477 and drawings of components as listed on DL-SC-A-595476.

3.5 Electrical installation.- Shelter Facility S-403( )/TSC-76.

3.5.1 All AC power wiring shall be in accordance with the model, and comply with the requirements of this specification.

3.5.1.1 Dielectric breakdown and insulation resistance, power wiring.- All AC power wiring shall have an insulation resistance of not less than 100 megohms when measured between:

a. "Hot line" and "Neutral"
b. "Neutral" and "Ground"
c. "Hot line" and "Ground"

Insulation resistance shall be measured no less than 10 seconds after a potential of 500 V DC is applied instantaneously, and with the voltage still applied. (See 4.6.1.8)

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3.5.2 Signal Wiring. The signal wiring shall be in accordance with the schematics provided, and the model, unless otherwise noted in the contract, and shall comply with the requirements of this specification.

3.5.2.1 Continuity, signal wiring.- There shall be electrical continuity between each terminal as shown on the drawings provided.

3.5.2.2 Cord assemblies.- Patch cords and air-conditioner cable assemblies shall be constructed in accordance with MIL-C-3885.

3.5.2.3 Dielectric breakdown and insulation resistance, signal wiring.- All signal wiring including individual wires, twisted pairs, cable harnesses, and cable assemblies shall have an insulation resistance of not less than 100 megohms when measured between each conductor and the remaining conductors, metallic connector shells, conduit, wireduct, and shelter ground. Insulation resistance shall be measured at least 10 seconds after an instantaneous potential of 500 V DC is applied. (See 4.6.2.3)

3.6 Military truck transport.- The shelter facility shall be capable, without any extra tying, taping, etc, of being transported over cross country terrain by military vehicle and shall withstand the test of 4.7. After testing per 4.7, there shall be no deformation, buckling, delamination, or damage to any part of the shelter facility.

3.7 Rail impact test.- The shelter facility shall be capable, without any extra tying, taping, etc, of being loaded, blocked and braced on a flatcar for shipment by rail, and shall withstand the test of 4.8. After each test there shall be no deformation, buckling, delamination or damage to any part of the shelter facility.

3.8 Drops, flat and rotational.- The shelter facility shall withstand, without any extra tying, taping, etc, drops of 18 inches onto concrete. After each of the drop tests of 4.9.1 and 4.9.2, there shall be no deformation, buckling, delamination or damage to any part of the shelter facility.

3.9 Watertightness.- The shelter facility shall withstand the test of 4.10. After the test, there shall be no leakage into the shelter facility or its walls, floor, roof, or doors.

3.10 Fording.- The shelter facility prepared for transportation on a military cargo truck, shall withstand the test of 4.11. After the test, there shall be no leakage into the shelter facility or its walls, floor, or doors.

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed the head shall not protrude more than .062 inch from the inside skin of the shelter. When installed each insert shall be capable of withstanding the tensile load and torque load referenced in paragraph 4.13. There shall be no damage to the insert threads,

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the insert shall not have rotated from its original position, and there shall be no distortion of the panel, mounting member, or insert.

3.12 Cleaning.-

3.12.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in an applicable portion of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.12.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.13 Finish, protective.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231.

3.14.2 Nameplates.- Each electrical equipment shelter facility S-403( )/TSC-76 shall be serial numbered and provided with a nameplate, located as on the model.

3.14.3 Aircraft loading data plate.- The data plate for the S-280/G Shelter shall be covered with the data plate of the shelter facility.

3.14.4 Interior and exterior marking.- Interior and exterior markings shall be as depicted in the model.

3.15 Riveting.- Riveted joints shall be tight, the joined points shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Soldering.-

3.16.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections but no acid or acid salts shall be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above they shall be completely neutralized and removed immediately after use.

3.16.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other

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foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heat during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.17 Tropicalization of material.- Canvas and materials as indicated in the model shall be treated to be resistant to moisture and fungi as follows:

3.17.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.17.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished item when used for the purpose intended.

3.17.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements.

3.17.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.1)

3.18 Welding.- Welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Finished welds shall not be ground. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be complete neutralized and removed. (See 4.4)

3.19 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.19.1 Cabling.- Insulated wires shall be formed into cables as indicated on drawings. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.19.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for at least two replacements of the part to which the wire or cable is connected.

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3.19.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.19.4 Splicing.- Wires shall not be spliced.

3.19.5 Connections.-

3.19.5.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.19.5.2 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.12)

3.20 Seal, vapor.- All joints and raw edges of the framing for the blower exhaust, power and signal entrance boxes, and air-conditioner cutouts shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall have the properties of 3.20.1, 3.20.2, and 3.20.3. (See 4.14)

3.20.1 Sealer shear at temperature.- The minimum shear strength of the fully cored sealer and its bond to the aluminum shall be as follows:

200 psi at ambient (80°F ±15°)
200 psi at -65°F
75 psi at 200°F

3.20.2 Sealer shear after humidity exposure.- The sealer shall withstand exposure to 100 per cent relative humidity at 160°F for 2 weeks and have a shear strength of 200 psi minimum.

3.20.3 Sealer shear after salt spray exposure.-After 2 weeks exposure to 20 per cent Na Cl solution at 95°F the sealer shall have a shear strength of 200 psi minimum.

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3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts, such as mounts, brackets, cabinets, etc, shall be physically and functionally interchangeable, without modification of such items, or of the shelter, or of the mounting locations. (See 4.15) Individual items shall not be hand-picked for performance or to fit the assigned position, mount or bracket. Reliance shall not be placed on an unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with applicable portions of the following paragraphs:

3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Soldering
3.17 Tropicalization of material
3.18 Welding
3.19 Wiring and Cabling
3.20 Seal, Vapor

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be as classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.17)

4.3 Preproduction inspection.- This inspection shall be performed by the contractor on the first article unless otherwise specified in the contract. It shall consist of the inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified

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for Group A, Group B, and Group C (see tables II, III and IV respectively). The preproduction inspection shall be performed in this order: (a) Mounting inserts, (b) Power wiring, (c) Signal wiring, (d) Military truck, (e) Rail transport, (f) Flat drop, (g) Rotational drop, (h) Watertightness, (i) Fording. The last tests performed shall be a repeat of the electrical tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Req Para	Insp Para
Military truck transportation	3.6	4.7
Rail transportation	3.7	4.8
Sealer, shear	3.20	4.14

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Finish	3.13
Marking	3.14
Welding	3.18

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 for 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5) Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the government.

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Table II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.22	4.16	1.5 dphu**	6.0 dphu
Electrical				*
Power wiring	3.5.1	4.6.1		
Operation of distribution panel circuits	3.5.1.1	4.6.1.1 through 4.6.1.8	1.0% for the entire group of power and signal wiring	
Dielectric breakdown and Insulation resistance, Power wiring	3.5.1.1	4.6.1.9		
Signal wiring	3.5.2	4.6.2		
Distribution and continuity	3.5.2.1	4.6.2.1		
Patch cords	3.5.2.2	4.6.2.2		
Dielectric Breakdown	3.5.2.3	4.6.2.3		

* All electrical defects are considered major
** dphu - defects per hundred units

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for AQL's listed in Table III shall be as follows:

AQL	Inspection level for normal and tightened and reduced inspection
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in the order listed in Table III.

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Table III - Group B inspection

Inspection	Rqr Para	Insp Para	AQL (see 4.5.2.1)
Insert test	3.11	4.13	4.1%
Watertightness	3.9	4.10)	(6.5%
Fording	3.10	4.11)	(for group
Interchangeability	3.21	4.5	4.0%

4.5.3 Group C inspection.- This inspection shall consist of the tests specified in Table IV and shall be performed on sample units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

Table IV - Group C inspection

Inspection	Rqr Para	Insp Para
Solderless-terminal-lug-connection	3.19.5.2	4.12
Flat drop	3.8	4.9.1
Rotational drop	3.8	4.9.2

4.5.3.1 Sampling for inspection.- One sample shall be selected at random for each group of inspection in Table IV from the first 50 units, or fraction thereof, produced and one sample from each additional 150 units and fraction thereof produced.

4.5.3.2 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspection which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Electrical installation tests.- The tests of 4.6.1 and 4.6.2 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115 VAC, 60 cycles, single phase.

4.6.1 Power wiring facility.- The power wiring shall be inspected for conformance with the model as furnished.

4.6.1.1 Test of main breaker circuit.-

a. Remove neon power indicator lamp. Switch main breaker of OFF.

b. Using a low voltage indication device (such as a buzzer) verify continuity between terminals 1 of one Power Receptacle U-238( )/G and terminal 1 of other U-238( )/G in Power Entrance Box.

c. Repeat b for terminals 2 and 4 of both receptacles.

d. Install neon power indicator lamp.

e. Connect a 115 VAC 60 cycle power source to the facility through one of the U-238( )/G power receptacle. The neon power indicator lamp shall be lighted, and the voltmeter shall indicate a nominal 115 VAC.

4.6.1.2 Test of breaker No 1 circuit.-

a. Switch on breaker No 1, neon indicator shall light.

b. Turn on "blackout by pass" switch.

c. Turn on "cold start" lamp switch. Incandescent lamps shall light.

d. Turn on light switch. Fluorescent lamps shall light.

e. Close door and turn off blackout bypass switch. All lamps shall remain lighted.

f. Unlatch door. All lamps shall extinguish before latch is in a full open position. Neon power indicator lamp shall remain lighted.

g. Relatch door. All lamps shall be lighted when latch is in the locked position.

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h. Repeat f and g twice.

i. Turn on blackout bypass switch and leave on for remainder of inspection. All lamps shall be lighted.

j. Switch breaker No 1 OFF and then ON. All lamps shall alternately extinguish and relight.

4.6.1.3 Test of breaker No 2 circuit.-

a. Switch on breaker No 2, neon indicator shall light. (Check that vent cover is open). Plug in blower.

b. Turn on switch for blower. Blower shall operate and shall exhaust air from shelter.

c. Turn off switch for blower. Blower shall stop.

4.6.1.4 Test of breaker No 3 circuit.-

a. Switch on breaker No 3, neon indicator shall light. Plug electric heater into heater receptacle.

b. Set switch on heater to HEAT position and turn thermostat to temperature higher than ambient. Heater shall blow warm air after a few minutes of operation.

c. Turn thermostat down to a temperature colder than ambient. Heating coil shall turn off indicated by a drop in the ammeter reading. Fan shall continue to blow air.

d. Set switch to a straight up position. Heater shall turn off.

e. Set switch to FAN position. Heater shall blow air.

f. Switch off breaker No 3. Heater shall turn off.

4.6.1.5 Test of breaker No 4 circuit.-

a. Switch on breaker No 4, neon indicator shall light.

b. Plug 115 VAC test lamp into each side of all "convenience" duplex receptacles. Test lamp shall light in each position.

c. Switch off breaker No 4. The test lamp shall not light when plugged into any of the "convenience" receptacles.

4.6.1.6 Test of breaker No 5 circuit.-

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a. Switch on breaker No 5, neon indicator shall light.

b. Plug 115 VAC test lamp into each equipment receptacle.

c. Switch off breaker No 5. The test lamp shall not light when plugged into any equipment receptacle.

4.6.1.7 Test of breaker No 6 circuit.-

a. Switch on breaker No 6, neon indicator shall light.

b. Connect air-conditioner power cable to receptacles on air-conditioner and shelter.

c. Turn air-conditioner fresh air switch to OPEN position and fan-cool switch to FAN. Air current shall emit from air-conditioner. Record current indicated on ammeter.

d. Turn fan-cool switch to COOL position and fresh air switch to CLOSED position. Turn thermostat control to lowest temperature. After a period of about 5 minutes the temperature of the air from the air-conditioner shall be cooler than the ambient air in the shelter. The unit shall function smoothly and the ammeter reading shall have increased above the first observed reading.

e. Turn off breaker No 6 air-conditioner shall stop functioning. Return fan-cool switch to OFF position.

4.6.1.8 Dielectric breakdown and insulation resistance, power wiring.- The dielectric breakdown and insulation resistance of the power wiring shall be tested for conformance with 3.5.1.1 as described below.

a. Remove all fluorescent lamps from their fixtures.

b. Remove all starters from fluorescent fixtures.

c. Remove cover from power distribution box. Disconnect voltmeter from circuit. Disconnect all green (ground) wires from neutral bux bar in power distribution box. Switch on all circuit breakers and switches.

d. Remove all incandescent and neon lamps.

e. Unplug all power cords (blowers, air-conditioners, heaters, intercom, radio, power supply etc) from receptacles in wireduct.

f. Apply an instantaneous voltage of 500 volts DC between terminal 1 of Power Receptacle U-238( )/G and terminal 2 of the same receptacle. After 10 seconds measure the resistance between terminal 1 and terminal 2. Resistance shall not be less than 100 megohms. CAUTION. After each test, discharge the conductor under test to ground to eliminate danger of shock.

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g. Repeat f replacing terminal 2 (2 places) with terminal 4.

h. Repeat f replacing terminal 1 (2 places) with terminal 4.

i. Restore shelter facility to original condition; that is, replace all lamps, cover, wires, etc.

4.6.2 Signal wiring, facility.- The signal wiring installation shall be inspected for conformance with the model. Jack designations (such as A16J24 etc) are as follows:

a. The first letter defines which panel the particular jack is located. A is the panel on the left, B is the center panel, and C is the panel on the right. (With the viewer facing the jack panels.)

b. The digit or digits following is the number of the jack strips, numbering consecutively from the top, starting with (1).

c. The digit or digits following the (j) indicate the jack numbers, with the numbering from left to right, starting with (1).

4.6.2.1 Continuity tests.-

4.6.2.1.1 Fabrication of test fixtures.- Test fixtures shall be fabricated in the following manner, and shall hereafter be identified by the referenced title.

a. Fixture No 1 - Jumper together all terminals of a J-1077 junction box. Connect one end of CX-4556/C Cable to box.

b. Fixture No 2 - Same as a above.

c. Fixture No 3 - Connect 26 phone plugs to bare end of 15 foot stub signal cable. Solder tips of plugs to "A" terminals, and rings to "B" terminals. (The plugs should be numbered from 1 to 26 and connected to the corresponding pins of the connector. Mate connector end of cable to J-1077 box prepared as in a above.

d. Fixture No 4 - Same as c above.

e. Fixture No 5 - Stub signal cable with no terminations on bare end other than tinned leads, connector end terminated in Junction box as in a above.

f. Shorted plug - Connect a diode across the tip and ring of a phone plug, conducting from tip to ring.

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g. Reverse shorted plug. - Connect a diode across the tip and ring of a phone plug conducting from ring to tip.

h. Dummy plug. - A phone plug with discontinuity between tip and ring.

i. Continuity tester.- A low voltage continuity indicated device, in series with a battery power supply, connected to a phone plug. The positive side of the unit shall be connected to the tip of the phone plug, and the negative side to the ring.

j. Fixture No 6. Place jumper wires on a J-1077 junction box paralleling all the "A" pins. Repeat for all "B" pins. Mate connector of CX-4566/C Cable to one connector of box. Mate connector of another CX-4566/C Cable to other connector of box.

k. Shorted connector. A mate to U-187( )/G Connector, with all pins shorted together. (48 each will be required).

l. Fixture No 7. Same as c above.

4.6.2.1.2 Continuity tests.-

a. Switched pairs.-

(1) With continuity tester, verify continuity between pin 26A of connector J1A and tip of A29J15.

(2) Connect one end of fixture No 6 to connector J1A of shelter.

(3) Connect the other end of fixture No 6 to connector J1B of shelter.

(4) Insert continuity tester into A29J15. Test shall indicate open circuit.

(5) Insert shorting plug into all jacks on jack strips A1 through A4 in sequence. Continuity shall be indicated for all insertations.

(6) Repeat steps (3) through (5) for:

(a) Connectors J2A and J2B and jack strips through A5 to A8.

(b) Connectors J3A and J3B and jack strips through A9 to A12.

(c) Connectors J4A and J4B and jack strips through A13 to A16.

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(d) Connectors J5A and J5B and jack strips thru A17 to A20

(e) Connectors J6A and J6B and jack strips thru A21 to A2

(f) Connectors J7A and J7B and jack strips thru A25 to A28

(g) Connectors J8A and J8B and jack strips thru B1 to B4

(h) Connectors J9A and J9B and jack strips thru B5 to B8

(i) Connectors J10A and J10B and jack strips thru B9 to B12.

(j) Connectors J11A and J11B and jack strips thru B13 to B16.

(k) Connectors J12A and J12B and jack strips B17 thru B20

(l) Connectors J13A and J13B for jack strips B21 thru 24

(m) Connectors J14A and J14B for jack strips B25 thru 28

(n) Connectors J15A and J15B for jack strips B29 thru 32

(o) Connectors J16A and J16B for jack strips C1 thru C4

(p) Connectors J17A and J17B for jack strips C5 thru 8

(q) Connectors J18A and J18B for jack strips C9 thru 12

(r) Connectors J19A and J19B for jack strips C13 thru 16

(s) Connectors J20A and J20B for jack strips C17 thru 20

(t) Connectors J21A and J21B for jack strips C21 thru 24

(u) Connectors J22A and J22B for jack strips C25 thru 28

b. Direct Continuity.-

(1) Insert continuity tester into A3J1. Test shall indicate open circuit.

(2) Insert shorting plug into A4J1. Test shall indicate open circuit.

(3) Insert dummy plug into A1J1. Test shall indicate continuity. Remove plug and open circuit shall be indicated.

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(4) Insert dummy plug into A2J1. Test shall indicate continuity. Remove plug and open circuit shall be indicated.

(5) Repeat steps (1) through (4) for jacks 2 through 24 on strips A1 to A4.

(6) Insert continuity tester into A29J15. Test shall indicate open circuit.

(7) Insert shorting plug into A4J26. Test shall indicate continuity.

(8) Insert dummy plug into A3J26. Test shall indicate open circuit. Remove dummy plug and continuity shall be indicated.

(9) Remove shorting plug from A3J26 and insert into A2J26. Continuity shall be indicated.

(10) Insert dummy plug into A1J26. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(11) Connect one end of CX-4566/C Cable to connector J23A, and the other end to connector J24A. This cable will remain connected for the duration of the continuity test.

(12) Connect one end of CX-4566 Cable to connector J1A, and the other end to J1B.

(13) Insert continuity tester into A3J25. Test shall indicate open circuit.

(14) Insert shorting plug into A4J25. Test shall indicate continuity.

(15) Insert dummy plug into A2J25. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(16) Insert dummy plug into A1J25. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(17) Insert continuity tester into A1J25. Test shall indicate open circuit.

(18) Insert shorted plug into A2J25. Test shall indicate continuity.

(19) Insert dummy plug into A3J25. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

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(20) Insert dummy plug into A4J25. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(21) Insert continuity tester into A29J15. Test shall indicate open circuit.

(22) Insert shorting plug into A4J26. Test shall indicate continuity.

(23) Insert dummy plug into A3J26. Test shall indicate continuity.

(24) Insert continuity tester into A3J1. Test shall indicate open circuit.

(25) Insert shorting plug into A4J1. Test shall indicate continuity.

(26) Insert dummy plug into A1J1. Test shall indicate continuity. Remove plug.

(27) Insert dummy plug into A2J1. Test shall indicate continuity. Remove plug.

(28) Repeat steps (24) through (27) for jack strips 2 to 24 and 26, on jack strips A1 through A4.

(29) Repeat steps (1) through (28) for connectors J2A to J7A, J2B to J7B with jack strips A5 to A28.

(30) Repeat steps (1) through (28) for connectors J8A to J15A and J8B and J15B with jack strips B1 to B23, with the substitution of A29J20 for A29J15 in step (21 .

(31) Repeat steps (1) through (28) for connectors J16A to J22A, and J16B to J22B with jack strips C1 to C28, with the substitution of C29J15 for A29J15 in step (21 .

(32) Insert continuity tester into A29J18. Test shall indicate open circuit.

(33) Insert shorted plug consecutively into A29J15 to A29J17. Test shall indicate continuity.

(34) Repeat steps (32) and (33) for A29J23 and A29J20 to A29J22, and C27J18 and C39J15 to C29J17.

(35) Insert continuity tester into A29J15. Test shall indicate open circuit.

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(36) Insert shorting plug into A29J20. Test shall indicate continuity.

(37) Insert dummy plug in sequence into A29J18, A29J23. Tests shall indicate open circuit. Remove plug.

(38) Remove shorting plug from A29J20 and insert into C29J15. Test shall indicate continuity.

(39) Repeat step (37) for A29J18 and C29J18.

(40) Remove continuity tester from A29J15, and insert into A29J20. Test shall indicate continuity.

(41) Repeat step (37) for J29J23 and C29J18.

(42) Connect continuity tester to tinned leads No 1 at SB-22 position (WHT-YEL). Test shall indicate open circuit.

(43) Short together SB-22 binding posts No 1 in signal entrance box. Test shall indicate continuity.

(44) Insert dummy plug into A29J5. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(45) Insert shorted plug into A29J5. Test shall indicate continuity.

(46) Remove jumper from binding posts. Test shall indicate continuity.

(47) Connect continuity tester to tip and ring of shorted plug. Continuity shall be indicated.

(48) Repeat steps (42) through (46) for the following SB-22 binding posts, and tinned leads at the SB-22 position.

(a) Binding posts No 2 and (WHT-ORN) tinned leads

(b) Binding posts No 3 and (WHT-BLK) tinned leads

(c) Binding posts No 4 and (WHT-RED) tinned leads

(d) Binding posts No 5 and (WHT-LT-BRN) tinned leads

(e) Binding posts No 6 and (WHT-DK-BRN) tinned leads

(f) Binding posts No 7 and (WHT-SIL) tinned leads

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(g) Binding posts No 8 and (WHT-DK-GRN) tinned leads

(h) Binding posts No 9 and (WHT-LT-GRN) tinned leads

(i) Binding posts No 10 and (WHT-VIO) tinned leads

(j) Binding posts No 11 and (WHT-SL) tinned leads

(k) Binding posts No 12 and (WHT-LT-BLU) tinned leads

(49) Insert continuity tester into A29J13. Test shall indicate open circuit.

(50) Short together binding posts 1 under "jacks" in signal entrance box. Test shall indicate continuity.

(51) Repeat steps (49) and (50) for:

(a) Binding posts '2' and A29J14

(b) Binding posts '3' and C29J13

(c) Binding posts '4' and C29J14

(52) Insert continuity tester into A29J1. Test shall indicate open circuit.

(53) Insert shorted plug into C29J1. Test shall indicate continuity.

(54) Repeat steps (52) and (53) for:

(a) A29J2 and C29J2

(b) A29J3 and C29J3

(c) A29J4 and C29J4

(55) Connect continuity tester to binding posts 'A' on binding posts panel. Test shall indicate open circuit.

(56) Short out binding posts 'A' in signal entrance box. Test shall indicate continuity.

(57) Repeat step (56) for binding posts 'B' on binding post panel, and in signal entrance box.

(58) Insert ohmeter plug into A30J1. Resistance shall be 35 $\pm$ 2 ohms.

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(59) Insert ohmmeter plug into A30J2. Resistance shall be 332 ± 2 ohms.

(60) Insert ohmmeter plug into A30J3. Resistance shall be 332 ± 2 ohms.

(61) Repeat steps (58) through (60) respectively for:

(a) A30J4, A30J5, A30J6

(b) A30J7, A30J8, A30J9

(c) A30J10, A30J11, A30J12

(d) A30J13, A30J14, A30J15

(e) A30J16, A30J17, A30J18

(f) C30J1, C30J2, C30J3

(g) C30J4. C30J5, C40J6

(h) C30J7, C30J8, C30J9

(i) C30J10, C30J11, C30J12

(j) C30J13, C30J14, C30J15

(k) C30J16, C30J17, C30J18

(62) Insert continuity tester into A31J1. Test shall indicate continuity.

(63) Insert dummy plug in sequence into A31J2, A31J3, and A31J4. Tests shall indicate open circuit for each insertion.

(64) Insert shorted plug, in sequence into A31J2, A31J3, and A31J4. Tests shall indicate continuity for each insertion.

(65) Insert continuity tester into A31J4. Test shall indicate continuity.

(66) Insert dummy plug in A31J1. Test shall indicate open circuit.

(67) Insert shorted plug in A31J1. Test shall indicate continuity.

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(68) Repeat steps (62) through (67) for the following groups of jacks.

(a) A31J5 through A31J8

(b) A31J9 through A31J12

(c) A31J13 through A31J16

(d) A31J17 through A31J20

(e) A31J21 through A31J24

(f) C31J1 through C31J4

(g) C31J5 through C31J8

(h) C31J9 through C31J12

(i) C31J13 through C31J16

(j) C31J17 through C31J20

(k) C31J21 through C31J24

(69) Insert continuity tester into A32J1. Test shall indicate open circuit.

(70) Insert shorted plug, in sequence into A32J2, A32J3, and A32J4. Tests shall indicate continuity for all insertations.

(71) Repeat steps (69) and (70) for the following groups of jacks:

(a) A32J5 through A32J8

(b) A32J9 through A32J12

(c) A32J13 through A32J16

(d) A32J17 through A32J20

(e) A32J21 through A32J24

(f) C32J1 through C32J4

(g) C32J5 through C32J8

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(h) C32J9 through C32J12

(i) C32J13 through C32J16

(j) C32J17 through C32J20

(k) C32J21 through C32J24

(72) Insert continuity tester into A29J11. Test shall indicate open circuit.

(73) Insert shorted plug into "4WS-2W BLACK" jack. Test shall indicate continuity.

(74) Insert continuity tester into A29J12. Test shall indicate open circuit.

(75) Insert shorted plug into "4WR-RED" jack. Test shall indicate continuity.

4.6.2.2 Patch cords.- Using the continuity tester, test between the tips of each plug on each patch cord, between the sleeves of each plug on each patch cord, and between one sleeve and one tip of each patch cord. Continuity shall be indicated in each of the first two tests, and discontinuity shall be indicated in the third test. (3.5.2.2)

4.6.2.3 Dielectric breakdown and insulation resistance, signal wiring.- The dielectric breakdown and insulation resistance of the signal wiring shall be tested for conformance with 3.5.2.3 (with either manual or automatic testing equipment) as described below:

a. Remove all telephone, teletype, patch cord, and intercom plugs from jacks in wireduct and patch panel. Check that lines are disconnected from SB-22 and TH-22's, and are not shorted to each other, the shelter, or any other metallic object.

b. Connect fixture No 1 to connector J1A on shelter.

c. Apply instantaneously a potential of 500 VDC between the binding posts of the fixture, and the shelter ground, and hold this potential for at least 10 seconds.

d. Immediately after performing step c and with the full potential still applied, measure the insulation resistance. The resistance shall not be less than 100 megohms.

e. Repeat steps b through d for connectors J2A to J24A, J23B and J24B.

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f. Insert plugs of fixture No 3 into A29J11, A29J12, A31J1, A31J5, A31J9, A31J13, A31J17, A31J21, C31J1, C31J5, C31J9, C31J13, C31J17, C31J21, A32J1, A32J5, A32J9, A32J13, A32J17, A32J21, C32J1, C32J5, C32J9, C32J13, C32J17, C32J21.

g. Repeat steps c and d.

h. Connect the tinned leads of fixture No 5 to all 18 pairs of binding posts in the signal entrance box, and to the 4 binding posts in the power entrance box. (NOTE. The above connections will hereafter be referred to as "Group 1").

i. Repeat steps c and d.

j. Remove the jumper between binding posts 1L and 2L in fixture junction box. (NOTE. Hereafter the binding posts in all fixtures will be referred to with two identifying characters. The first number will indicate the number marked in the junction box, and the second character, either 'R' or 'L' will indicate to the right or left of the number marked in the box.

k. Repeat steps c and d with the potential applied between 1L and 14R.

l. Shift the jumper from between 2L and 3L to between 1L and 2L.

m. Repeat step k.

n. Repeat steps l and m in sequence shifing the jumper from betwe 3L and 4L to between 2L and 3L, from between 4L and 5L to between 3L and 4L etc, in sequence through 13L, and from 13R through 1R, from 14L through 26L and from 26R through 14R.

o. Insert the plugs of fixture No 3 into the following jacks: A31J A31J3, A31J5, A31J7, A31J9, A31J11, A31J13, A31J15, A31J17,A31J19, A31J21, A31J23, A32J1, A32J5, A32J9, A32J13, A32J17, A32J21, A29J11, A29J12. (NOTE. Hereafter the referenced connections will be referred to as "Group 2").

p. Insert dummy plugs into the following jacks: A31J2, A31J4, A31J6, A31J8, A31J10, A31J12, A31J14, A31J16, A31J18, A31J20, A31J22, A31J24.

q. Repeat steps j through n.

r. Insert the plugs of fixture No 3 into the following jacks: C31J1, C31J3, C31J5, C31J7, C31J9, C31J11, C31J13, C31J15, C31J17, C31J19, C31J21, C31J23, C32J1, C32J5, C32J9, C32J13, C32J17, C32J21. (NOTE. Hereafter the referenced connections shall be referred to as "Group 3").

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s. Insert dummy plugs into the following jacks: C31J2, C31J4, C31J6, C31J8, C31J10, C31J12, C31J14, C31J16, C31J18, C31J20, C31J22, C31J24.

t. Repeat steps j through n.

u. Insert dummy plugs into all jacks on strip A2.

v. Connect fixture No 1 to connector J1A.

w. Repeat steps j through n.

x. Repeat step w, moving fixture No 1 and the dummy plugs to the following connectors, jack strips and single jacks:

(1) Connector J1B and jack strip A1 and A4J25

(2) Connector J2A and jack strip A6.

(3) Connector J2B and jack strip A5 and A8J25

(4) Connector J3A and jack strip A10.

(5) Connector J3B and jack strip A9 and A12J25

(6) Connector J4A and jack strip A14

(7) Connector J4B and jack strip A13 and A16J25

(8) Connector J5A and jack strip A18

(9) Connector J5B and jack strip A17 and A20J25

(10) Connector J6A and jack strip A22

(11) Connector J6B and jack strip A21 and A24J25

(12) Connector J7A and jack strip A26

(13) Connector J7B and jack strip A25 and A28J25

(14) Connector J8A and jack strip B2

(15) Connector J8B and jack strip B1 and B4J25

(16) Connector J9A and jack strip B6

(17) Connector J9B and jack strip B5 and B8J25

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(18) Connector J10A and jack strip B10

(19) Connector J10B and jack strip B9 and B12J25

(20) Connector J11A and jack strip B14

(21) Connector J11B and jack strip B13 and B16J25

(22) Connector J12A and jack strip B18

(23) Connector J12B and jack strip B17 and B20J25

(24) Connector J13A and jack strip B22

(25) Connector J13B and jack strip B21 and B24J25

(26) Connector J14A and jack strip B26

(27) Connector J14B and jack strip B25 and B28J25

(28) Connector J15A and jack strip B3C

(29) Connector J15B and jack strip B29 and B32J25

(30) Connector J16A and jack strip C2

(31) Connector J16B and jack strip C1 and C4J25

(32) Connector J17A and jack strip C6

(33) Connector J17B and jack strip C5 and C8J25

(34) Connector J18A and jack strip C10

(35) Connector J18B and jack strip C9 and C12J25

(36) Connector J19A and jack strip C14

(37) Connector J19B and jack strip C13 and C16J25

(38) Connector J20A and jack strip C18

(39) Connector J20B and jack strip C17 and C20J25

(40) Connector J21A and jack strip C22

(41) Connector J21B and jack strip C21 and C24J25

(42) Connector J22A and jack strip C26

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(43) Connector J22B and jack strip C25 and C28J25

(44) Connector J23A - see Note 1

(45) Connector J23B

(46) Connector J24A - see Note 2

(47) Connector J24B

NOTE 1. Insert dummy plugs into all J25 jacks on jack strips - A1, A5, A9, A13, A17, A21, A25, B1, B5, B9, B13, B17, B21, B25, B29, C1, C5, C9, C13, C17, C21, C25. Remove after test.

NOTE 2. Insert dummy plugs into all J25 jacks on the following jack strips - A3, A7, A11, A15, A19, A23, A27, B3, B7, B11, B15, B19, B23, B27, B31, C3, C7, C11, C15, C19, C23, C27. Remove after test.

y. Place shorted connectors on each signal connector of shelter facility.

z. Insert the plugs of fixture No 3 into the jacks of "Group 2".

aa. Insert the plugs of fixture No 4 into the jacks of "Groups 3".

ab. Connect the leads of fixture No 5 to"Group 1".

ac. Run patch cables between the following jacks:

(1) A4J1 to A7J1

(2) A8J1 to A11J1

(3) A12J1 to A15J1

(4) A16J1 to A19J1

(5) A20J1 to A23J1

(6) A24J1 to A27J1

(7) A28J1 to B3J1

(8) B4J1 to B7J1

(9) B8J1 to B11J1

(10) B12J1 to B15J1

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(11) B16J1 to B19J1

(12) B20J1 to B23J1

(13) B24J1 to B27J1

(14) B28J1 to B31J1

(15) B32J1 to C3J1

(16) C4J1 to C7J1

(17) C8J1 to C11J1

(18) C12J1 to C15J1

(19) C16J1 to C19J1

(20) C20J1 to C23J1

(21) C24J1 to C27J1

(22) C28J1 to A6J25

(23) A2J25 to A8J25

(24) A4J25 to C29J14

(25) C29J13 to A32J2

ad. Insert dummy plugs in all J25 jacks on all even numbered jack strips on all three jack panels except A2, A4, A6, A8, and in all J26 jacks on all odd numbered jack strips on all three jack panels.

ae. Repeat steps c and d with the potential applied between the tip of A3J1 and the tip of C32J2.

af. Move the patch cord plug from C29J13 to C32J1.

ag. Repeat step ae.

ah. Move the patch cord plug from A4J25 to A32J1.

ai. Repeat step ae.

aj. Move the patch cord plug from A2J25 to C29J13.

ak. Insert dummy plug into A2J25.

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al. Repeat step ae.

am. Remove dummy plugs of steps ad and ak with the exception of the plugs in the J26 jacks.

an. Insert dummy plugs into all J25 jacks on all odd numbered jack strips on all three jack panels.

ao. Repeat step ae.

ap. Move the patch cord plug from C28J1 to A4J5.

aq. Repeat step ae.

ar. Remove dummy plugs of step an.

as, Repeat steps ad and ak. These plugs will now remain in place for the duration of the test.

at. Repeat step ae.

au. Connect patch cord between C28J1 and A2J25.

av. Insert dummy plugs into all open jacks on jack strip C26.

aw. Repeat step ae.

ax. Remove the dummy plugs of step av.

ay. Repeat steps av through ax for the following jack strips: C25, C22, C21, C18, C17, C14, C13, C10, C9, C6, C5, C2, C1, B30, B29, B26, B25, B22, B21, B18, B17, B14, B13, B10, B9, B6, B5, B2, B1, A26, A25, A22, A21, A18, A17, A14, A13, A10, A9, A6, A5, A2, A1.

az. Remove the patch cord of step ac - 21.

ba. Repeat step ae.

bb. Replace the patch cord of step az.

bc. Repeat steps az through bb for all patch cords referenced in step ac - 1 through ac - 20.

4.7 Military truck transport.- The shelter facility with operational equipment (3.3) installed shall be transported on a government furnished M-715 truck over the specially prepared course descripted in paragraph 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph, and 10 laps at 20 mph. A lap is defined as traversing the course in one direction. The shelter facility shall meet the requirements of 3.6 after completion of the above test.

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4.7.1 Military truck transport course.- The course for the military truck transport test shall have twelve 4 inch by 8 inch timbers placed 25 feet apart on the 8 inch face and with the 4 inch face fully above the ground. The edges shall have a 1 inch by 1 inch bevel. The sixth and twelfth timbers shall be placed 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored securely by any convenient means. Ground on which timbers are placed shall be hard and level.

4.8 Rail transport, impact test.- The shelter facility with operational equipment (3.3) installed shall be loaded on a railroad flatcar (test car) in accordance with figure 1. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 pounds (impact car) and traveling at 9 mph (or equivalent impact energy conditions approved by the government) shall be impacted against the test car coupled to two other empty cars. These cars shall be stationary with all brakes off. Four impacts shall be performed, one impact into each end of the test car. The cabling and blocking holding the shelter facility in place shall be tight at the start of each impact run. If the cable or blocking is torn loose or if the load on the impact car shifts, the run shall be repeated. A damage survey shall be made after each impact and the shelter facility shall meet the requirements of 3.7.

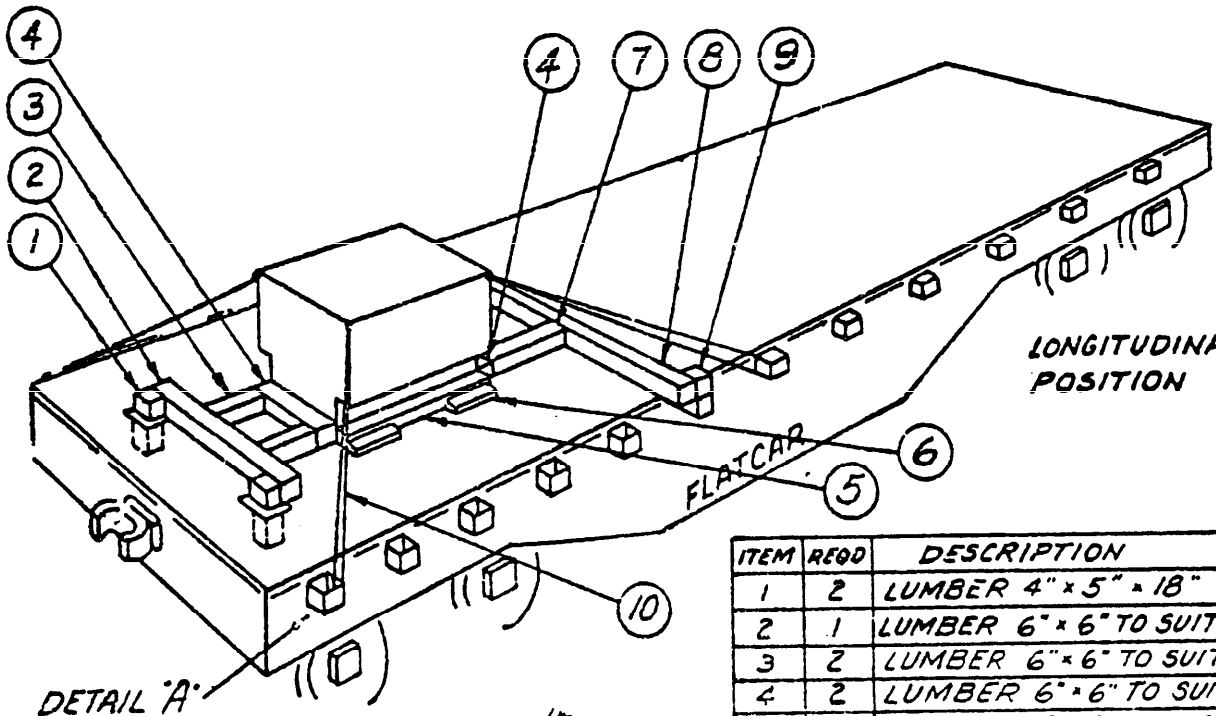
4.9 Drop tests.-

4.9.1 Flat drop.- The shelter facility with operational equipment (3.3) installed shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of this drop, the shelter facility shall meet the requirements of 3.8.

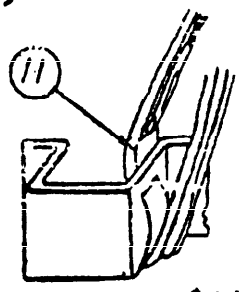
4.9.2 Rotational drop.- The shelter facility with operational equipment (3.3) installed shall have one edge placed on a nominal 4 inch high board. The opposite edge of the shelter facility shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the shelter facility. Upon completion of these drops, the shelter facility shall meet the requirements of 3.8. In each drop test, the 4 inch high member shall be placed under the skids and the eighteen inch measurement shall be taken from the bottom of the skids.

4.10 Watertightness test.- The shelter facility without the use of any additional caulking, sealing, taping, etc, shall be subjected to the following test: Each of the five exposed surfaces of the shelter, with covers opened, shall be sprayed with water from nozzles, model G29SQ (or CG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof

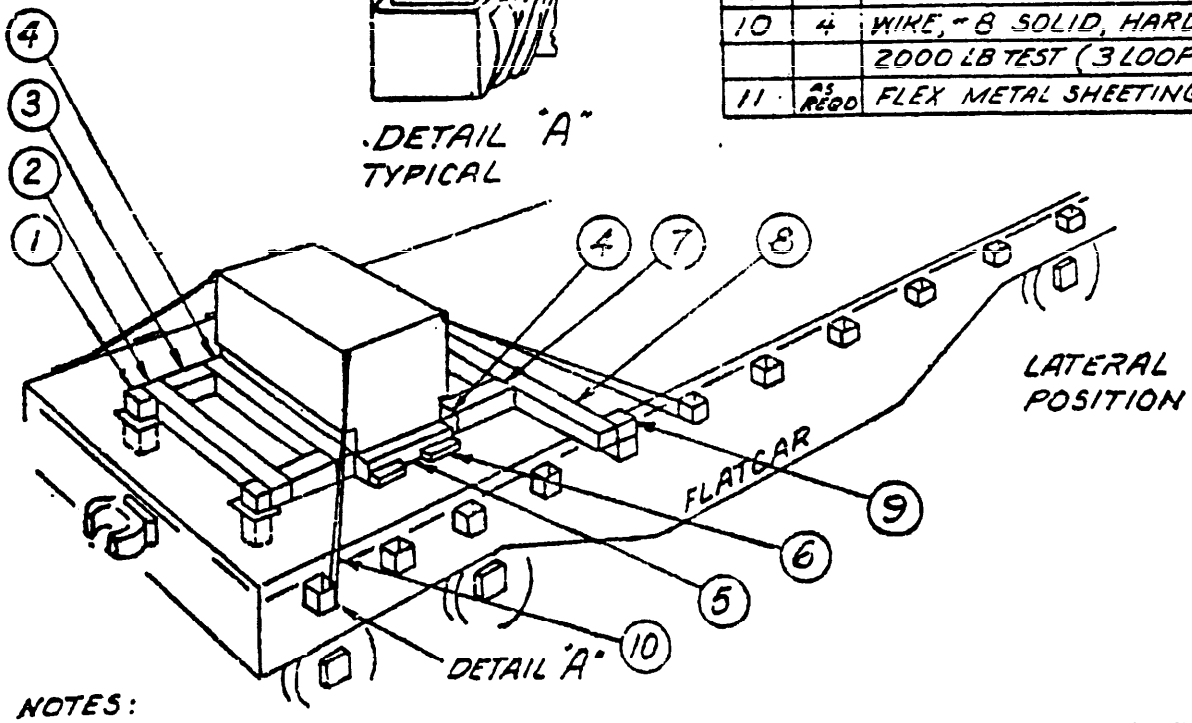
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ITEM	REQD	DESCRIPTION
1	2	LUMBER 4" x 5" x 18"
2	1	LUMBER 6" x 6" TO SUIT
3	2	LUMBER 6" x 6" TO SUIT
4	2	LUMBER 6" x 6" TO SUIT
5	2	LUMBER 4" x 4" TO SUIT
6	4	LUMBER 2" x 4" x 14"
7	2	LUMBER 6" x 6" TO SUIT
8	1	LUMBER 6" x 6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE, #8 SOLID, HARD 2000 LB TEST (3 LOOPS)
11	AS REQD	FLEX METAL SHEETING



DETAIL "A" TYPICAL



NOTES:

1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.

2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

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nozzles which shall be at a distance of thirty-two inches and the row of nozzles in line with the air-conditioner where the 19 inches shall be measured from the extreme vertical surface of the air-conditioner. (See figure 2 for nozzle location). Each panel shall be sprayed for a period of forty minutes. (If the watertightness test is to be followed by the fording test of 4.11, the bottom row of nozzles, with the exception of the nozzles facing the door-end panel, can be eliminated). Upon completion of the above test, the shelter facility shall meet the requirements oe 3.9. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the government at time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.11 Fording test.- The shelter facility, prepared for transportation on a military cargo truck and without the use of any additional sealer, caulking, taping, fording kit, etc, shall be immersed in water to a depth of thirty inches, measured from the bottom of the shelter skids, for a period of two hours. The shelter shall be restrained by external tiedowns. Upon completion of this test, the shelter facility shall meet the requirements of 3.10. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.12 Testing of solderless-terminal-lug connections. Connections made with solderless terminal-lugs shall be tested to meet the requirements of 3.19.5.2.

4.13 Insert test.- Two per cent of the total number of any size insert, but not less than five inserts, shall be randomly selected for test in each sample facility. Each selected insert shall have a torque load, in accordance with Table V applied to the bolt. (Note. Bolt shall hold a stationary plate against head of insert). After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of sample facility.

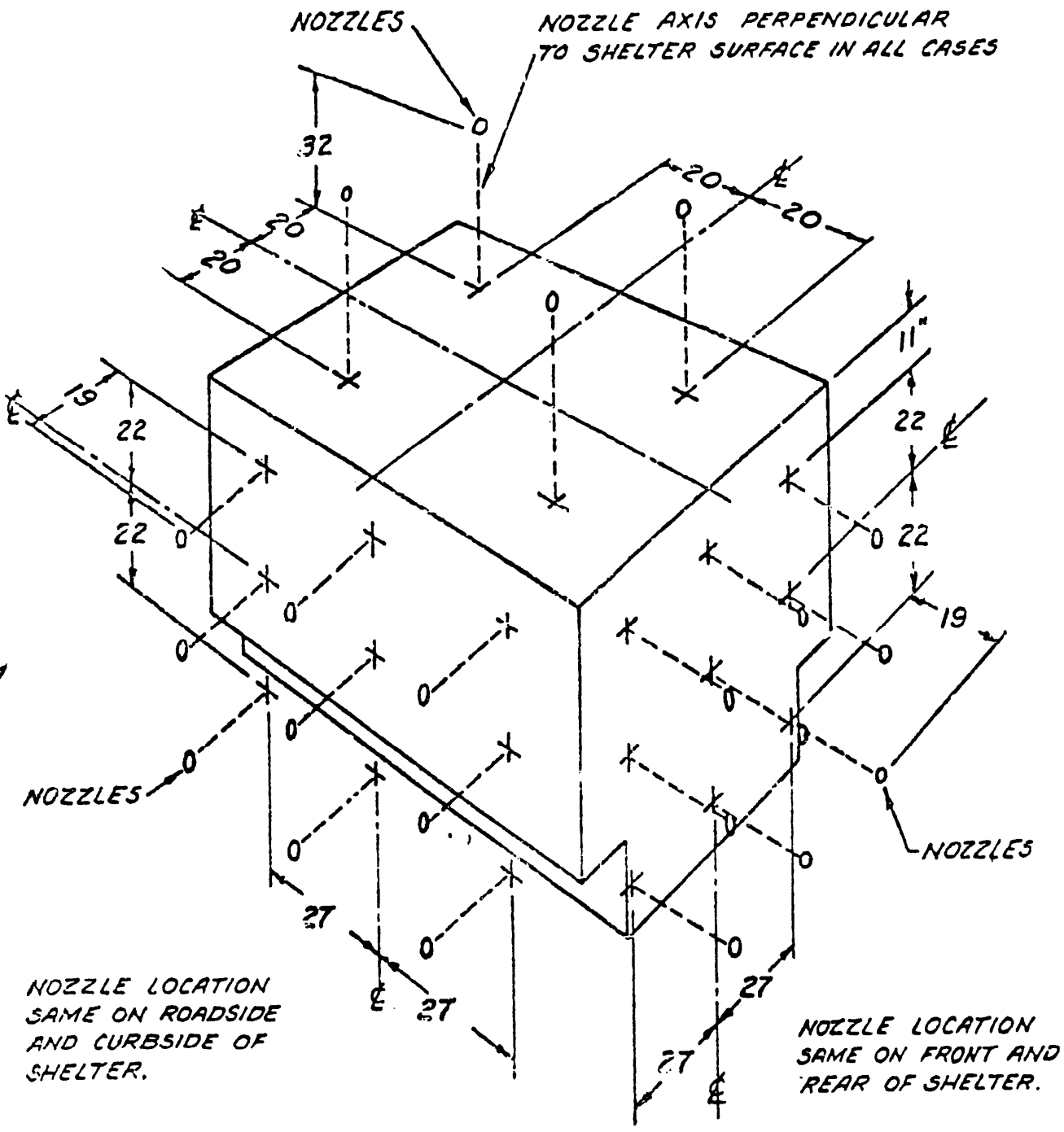
Table V - Production torque load

Insert size	Torque (in lbs)
5/16 - 18	100
1/4 - 20	60
10 - 32	30
8 - 32	20

4.14 Sealer shear tests.- A tensile pull test shall be performed on sample lap joints consisting of two (2) one-inch wide strips of aluminum (0.062 inch thick) overlapped one-half inch, giving contact area of one-half square inch and bonded with sealer material over no more than the specified one-half square inch area. The samples shall be tested over the conditions of 3.20.1 through 3.20.3. Five specimens shall be used for each test condition and tests shall be to destruction with rate of application of load constant at one inch per minute maximum.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

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NOTE:
GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.

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a. Location and size of mounting inserts.

b. Size and mounting dimensions of brackets.

c. Size of antenna pass-through plugs.

d. Size and location of radio shelf mounting nuts.

e. Size and mounting dimensions of blowers and heaters.

4.16 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting insert missing or mislocated. 2. Sealer improperly applied (holes, separations, or lack of adhesion). 3. Welds - cracked or porous. 4. Parts, brackets, or equipment loose or missing. 5. Drawers, door, etc inoperative. 6. Blower inoperative or rotating in wrong direction. 7. Locking or holding devices inoperative. 8. Lock washer or lock nut missing. 9. Burrs and sharp edges in wire duct not removed. 10. Incorrect or missing labels on binding parts and jacks.
Minor	1. Screw loose. 2. Drawers, doors, etc, difficult to operate. 3. Locking or holding devices difficult to operate. 4. Finish - abrasions or scratches. 5. Burrs and sharp edges on brackets not removed.

4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-55505.

4.18 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.1), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.16 Visual and Mechanical

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing of Shelter Facility S-403( )/TSC-76 shall be in accordance with the applicable requirements of Specification MIL-P-55505.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(c) The specific paragraphs of section 5 which are applicable to particular procurement.

(d) Preproduction pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection to be performed thereon - including rough handling test, which will not be performed as procurement inspection.

(4) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. This data should be submitted to the contracting officer with the preproduction pack.

(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) List of each individual equipment item or equivalent dummy load, with details, to be used in testing the shelter facility (3.3).

(h) Submission of the statement of treatment referenced in 3.17.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Definitions.-

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6.2.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.2.2 Examination.- Examination consists of simple generally nondestructive determinations of compliance, without use of special testing equipment.

6.2.3 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: S403X/TSC-76. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian
Army - EL

Preparing Activity:
Army - EL

Project No 5805-A106

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 OCT 64 1426

ESC-FM 2011-66

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